

Kaiwen Wu

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Research

I am interested in both theory and applications of machine learning. In my PhD, I have been working on various areas in probabilistic machine learning including Gaussian processes, variational inference, Bayesian optimization, and active learning. Before starting my PhD, I worked on generative modeling, convex optimization, and adversarial robustness of deep learning models.

Education

University of Pennsylvania

Ph.D. in Computer and Information Science
Advisor: Jacob R. Gardner

Philadelphia, PA
2021 – Present

University of Waterloo

M.Math. in Computer Science (Thesis Option)
Advisor: Yaoliang Yu

Waterloo, ON
2018 – 2020

Nanjing University

B.Sc. in Computer Science

Nanjing, China
2014 – 2018

Work Experience

Meta

Research Scientist Intern

- Worked on Bayesian optimization for AutoML.

New York, NY
Jun 2025 – Oct 2025

Meta Reality Labs

Research Scientist Intern

- Worked on experimental designs for VR/AR hardware using active learning.

Redmond, WA
Aug 2024 – Dec 2024

Borealis AI

Research Intern

- Worked on robust mean estimation on noisy data using generative adversarial networks.
- Published a paper at AISTATS 2020.

Waterloo, ON
Jan 2019 – May 2019

Open Source Contributions

BoTorch (<https://github.com/pytorch/botorch>)

A Python package for Bayesian optimization in PyTorch

- Implemented an improved elliptical slice sampling method for truncated normal distributions.

Publications

See also at my Google Scholar .

Refereed Conference Proceedings

[C1] **Kaiwen Wu** and Jacob R. Gardner

Understanding Stochastic Natural Gradient Variational Inference
Proceedings of the 41st International Conference on Machine Learning (ICML). 2024.

[C2] **Kaiwen Wu**, Jonathan Wenger, Haydn T Jones, Geoff Pleiss, and Jacob Gardner

Large-Scale Gaussian Processes via Alternating Projection
Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS). 2024.

- [C3] Kyurae Kim, Jisu Oh, **Kaiwen Wu**, Yian Ma, and Jacob R. Gardner
On the Convergence of Black-Box Variational Inference
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- [C4] Natalie Maus, **Kaiwen Wu**, David Eriksson, and Jacob Gardner
Discovering Many Diverse Solutions with Bayesian Optimization
Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS). 2023.
- [C5] **Kaiwen Wu**, Kyurae Kim, Roman Garnett, and Jacob R. Gardner
The Behavior and Convergence of Local Bayesian Optimization
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- [C6] Quan Nguyen, **Kaiwen Wu**, Jacob Gardner, and Roman Garnett
Local Bayesian optimization via maximizing probability of descent
Advances in Neural Information Processing Systems (NeurIPS). 2022.
- [C7] **Kaiwen Wu**, Gavin Weiguang Ding, Ruitong Huang, and Yaoliang Yu
On Minimax Optimality of GANs for Robust Mean Estimation
Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS). 2020.
- [C8] **Kaiwen Wu**, Allen Wang, and Yaoliang Yu
Stronger and Faster Wasserstein Adversarial Attacks
Proceedings of the 37th International Conference on Machine Learning (ICML). 2020.

Workshop Papers

- [W1] **Kaiwen Wu** and Jacob R. Gardner
A Fast, Robust Elliptical Slice Sampling Method for Linearly Truncated Multivariate Normal Distributions
NeurIPS Workshop on Bayesian Decision-making and Uncertainty. 2024.
- [W2] Guojun Zhang, **Kaiwen Wu**, Pascal Poupart, and Yaoliang Yu
Newton-type Methods for Minimax Optimization
ICML Workshop on Beyond First-order Methods in Machine Learning Systems. 2021.

Teaching Experience

University of Pennsylvania

- Teaching Assistant, CIS 520 Machine Learning Fall 2023
- Teaching Assistant, ESE 204 Decision Models Fall 2022

University of Waterloo

- Teaching Assistant, CS 370 Numerical Computation Spring 2020
- Teaching Assistant, CS 486/686 Introduction to Artificial Intelligence Spring 2019
- Teaching Assistant, CS 480/680 Introduction to Machine Learning Fall 2018

Honors & Awards

- NeurIPS Top Reviewer 2024
- Vector Scholarship in Artificial Intelligence, Vector Institute 2018
- Entrance Scholarship, University of Waterloo 2018
- Jimin Liu Scholarship, Nanjing University 2017

Academic Service

Conference Reviews

- Program Committee Member, AAAI Conference on Artificial Intelligence (2021, 2026)
- Reviewer, Conference on Neural Information Processing Systems (2023 – 2025)
- Reviewer, International Conference on Artificial Intelligence and Statistics (2021, 2024)
- Reviewer, International Conference on Learning Representations (2024 – 2025)
- Reviewer, International Conference on Machine Learning (2023 – 2025)

Journal Reviews

- Reviewer, Transactions on Machine Learning Research (2025)
- Reviewer, Journal of Machine Learning Research (2025)